Category Theory for Mathematical Physics — Exercise Sheet 6

F-symbols and the pentagon

Variant: Solutions

Conventions as in Lecture 6: the categories are multiplicity-free, with vertices ψ_c^{ab}; the F-matrix F_d^{abc} has rows e and columns f in the order of the list of simple objects (Definition 6.1); an F-symbol with a unit leg is 1 when its trees exist, and every symbol whose trees do not exist is 0. Fibonacci has $1, \tau$ and Ising $1, \sigma, \psi$ (the bare ψ is Ising's fermion; a vertex always carries its labels). $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, as in Lecture 3. The gauge numbers are u_c^{ab} (Theorem 6.4); $\varphi = (1+\sqrt{5})/2$; H is the Hadamard matrix; x, A, B, C, D are the Fibonacci unknowns of Theorem 6.6. In Exercise 5, $\uparrow, \downarrow$ are the spin-$\frac{1}{2}$ states $m = \pm\frac{1}{2}$. In Exercise 6, $\gamma_1, \gamma_2, \gamma_3$ name three F-symbols (the labels g of the pentagon and the group elements g of Lecture 3 are other letters).

Exercise 1 (Boson and semion). Take the fusion ring of $\mathbb{Z}_2$: $s \otimes s \cong 1$.

(i) Show that the only F-symbol without a unit leg is $x = [F_s^{sss}]_{11}$, and that the pentagon equation of Theorem 6.3 reduces to $x^2 = 1$.

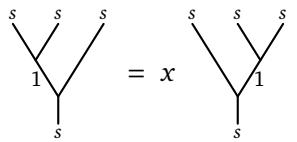
(ii) Show that x is gauge invariant, and conclude that the fusion ring has exactly two solutions up to gauge. Which categories of Lecture 3 are they, and what is κ_s for each?

(iii) Draw the F-move Equation (6.2) for $g = h = k = s$.

Solution. (i) Every product of simple objects is simple, so in $[F_d^{abc}]_{ef}$ the channels are forced: $e = ab$, $f = bc$, $d = abc$. Without a unit leg, $a = b = c = s$, so $e = f = 1$ and $d = s$: the only such symbol is $x = [F_s^{sss}]_{11}$. In the pentagon instance with $a = b = c = d = s$ the labels are forced too: $f = ab = 1$, $g = fc = s$, $l = cd = 1$, $k = bl = s$, $e = s^4 = 1$, and the sum has the single term $h = bc = 1$. The left side is $[F_1^{1ss}]_{s1}[F_1^{ss1}]_{1s} = 1$ (unit legs), and the right side is $[F_s^{sss}]_{11}[F_1^{s1s}]_{ss}[F_s^{sss}]_{11} = x \cdot 1 \cdot x$. So $x^2 = 1$. Every other instance has a unit among a, b, c, d, and the four cases are quick: the labels are again forced, and both sides become the same product. For $a = 1$, $b = c = d = s$, for instance, the left side is $[F_s^{sss}]_{11} \cdot 1 = x$ and the right side is $1 \cdot 1 \cdot [F_s^{sss}]_{11} = x$. So each other instance reads $1 = 1$ or $x = x$ (the script lists the distinct equations: only $x^2 = 1$).

(ii) By Theorem 6.4 the factor on $[F_s^{sss}]_{11}$ is $u_d^{af}\, u_f^{bc} / (u_e^{ab}\, u_d^{ec}) = u_s^{s1}\, u_1^{ss} / (u_1^{ss}\, u_s^{1s}) = 1$ (with $u_s^{1s} = u_s^{s1} = 1$). So no gauge transformation changes x, and the two solutions $x = \pm 1$ are not gauge equivalent: up to gauge there are exactly two. By Equation (6.2), $x = \omega(s,s,s)$: $x = 1$ is $\mathrm{Vec}_{\mathbb{Z}_2}$ and $x = -1$ the semion category of Example 3.9, as Proposition 3.11 found. With $d_s = 1$, Equation (6.4) gives $\kappa_s = x$: $+1$ for the boson and -1 for the semion.

(iii) The picture is Equation (6.2) with every leaf s, root s, and both inner edges labelled 1:



Exercise 2 (Fibonacci in terms of d_τ, and Yang–Lee). Let d_τ be a root of $d^2 = 1 + d$, and let $d_\tau^{-1/2}$ be a square root of d_τ^{-1}.

(i) Show that $x = 1$ and

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} d_\tau^{-1} & d_\tau^{-1/2} \\ d_\tau^{-1/2} & -d_\tau^{-1} \end{pmatrix}$$

satisfy the six equations listed before Theorem 6.6, and that $d_\tau = \varphi$ gives the unitary solution there.

(ii) For $d_\tau = \varphi$, write the inverse F-move Equation (6.3) for three τ's with total charge τ whose last two fuse to 1.

(iii) For $d_\tau = -1/\varphi$ (Yang–Lee) write the matrix, and show that it squares to the identity but is not unitary. Compute $\kappa_\tau = d_\tau A$ and $1 + d_\tau^2$ for both roots, and compare the second with $\dim \mathrm{Fib} = 2 + \varphi$ under $\sqrt{5} \mapsto -\sqrt{5}$. Why is Yang–Lee no anyon model?

Solution. (i) Write $d = d_\tau$. With $x = 1$, $A = d^{-1}$, $B = C = d^{-1/2}$ (so $BC = d^{-1}$) and $D = -d^{-1}$, the six equations before Theorem 6.6 read: $A = xBC$: $d^{-1} = d^{-1}$. $BC = x^2 A$: $d^{-1} = d^{-1}$. $xBC + A^2 = 1$: $d^{-1} + d^{-2} = 1$, which is $d^2 = 1 + d$ divided by d^2. $A^2 + BCD = 0$: $d^{-2} - d^{-2} = 0$. $BD = xBD$: true with $x = 1$. $x^2 = xD^2 + BC$: $1 = d^{-2} + d^{-1}$, as before. For $d = \varphi$, $A = \varphi^{-1}$ and $B = C = \varphi^{-1/2}$, the matrix of Theorem 6.6. (All the other pentagon equations hold as well; the script checks them, for both roots.)

(ii) By Equation (6.3), the right tree with channel $f = 1$ is $R_1 = \sum_e \overline{[F_\tau^{\tau\tau\tau}]_{e1}}\, L_e$, the first column conjugated: $R_1 = \varphi^{-1} L_1 + \varphi^{-1/2} L_\tau$. The matrix is real and symmetric, so the inverse move has the same numbers as the move itself.

(iii) For $d = -1/\varphi$: $d^{-1} = -\varphi$ and $d^{-1/2} = \sqrt{-\varphi} = i\sqrt{\varphi}$ (either root; the other is the gauge $q \mapsto iq$ of Theorem 6.6), so

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} -\varphi & i\sqrt{\varphi} \\ i\sqrt{\varphi} & \varphi \end{pmatrix}.$$

Its square has diagonal entries $\varphi^2 - \varphi = 1$ and off-diagonal entries $-\varphi\, i\sqrt{\varphi} + i\sqrt{\varphi}\, \varphi = 0$: $F^2 = 1$. It is not unitary: a unitary matrix has entries of modulus at most 1, and $\varphi > 1$. By Equation (6.4), $\kappa_\tau = d_\tau A = d_\tau d_\tau^{-1} = 1$ for both roots. And $1 + \varphi^2 = 2 + \varphi$, while $1 + \varphi^{-2} = 1 + (2 - \varphi) = 3 - \varphi$ (from $\varphi^{-1} = \varphi - 1$). Under $\sqrt{5} \mapsto -\sqrt{5}$, $\varphi \mapsto 1 - \varphi$, so $2 + \varphi \mapsto 3 - \varphi$: the global dimensions are Galois conjugates, as the categories are. An anyon model is a unitary fusion category (Lecture 5), whose F-matrices are unitary (Proposition 6.2); the Yang–Lee F-matrix is not, in any gauge with $|u| = 1$, and $A = -\varphi$ is gauge invariant with $|A| > 1$, so no gauge makes it unitary.

Exercise 3 (What a gauge can change). (i) For Fibonacci, with $u_1^{\tau\tau} = p$ and $u_\tau^{\tau\tau} = q$, compute from Theorem 6.4 the factors on x, A, B, C and D. Show that BC is invariant and B/C is multiplied by q^4/p^2. The numbers $x = 1$, $A = \varphi^{-1}$, $B = 1$, $C = \varphi^{-1}$, $D = -\varphi^{-1}$ solve the pentagon; can a gauge with $|u| = 1$ make them unitary?

(ii) For Ising, show that $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ is invariant, and compute $F_\sigma^{\sigma\sigma\sigma}$ and $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}$ after the gauge $u_\psi^{\sigma\sigma} = i$, all other u equal to 1. Is the new matrix unitary?

Solution. (i) For $[F_\tau^{\tau\tau\tau}]_{ef}$ the factor is $u_\tau^{\tau f} u_f^{\tau\tau} / (u_e^{\tau\tau} u_\tau^{e\tau})$. With $e = f = 1$: $u_\tau^{\tau 1} u_1^{\tau\tau} / (u_1^{\tau\tau} u_\tau^{1\tau}) = 1$ (A). With $e = f = \tau$: $q \cdot q / (q \cdot q) = 1$ (D). With $e = 1$, $f = \tau$: $u_\tau^{\tau\tau} u_\tau^{\tau\tau} / (u_1^{\tau\tau} u_\tau^{1\tau}) = q^2/p$ (B). With $e = \tau$, $f = 1$: p/q^2 (C). For $x = [F_1^{\tau\tau\tau}]_{\tau\tau}$ the factor is $u_1^{\tau\tau} u_\tau^{\tau\tau} / (u_\tau^{\tau\tau} u_1^{\tau\tau}) = 1$. So BC is invariant and B/C is multiplied by $(q^2/p)^2 = q^4/p^2$. The given numbers have $BC = \varphi^{-1} = A$ and solve the pentagon (they are the solution of Theorem 6.6 in the gauge $q^2/p = \varphi^{1/2}$). They are not unitary: the first row has $\varphi^{-2} + 1 > 1$. A gauge with $|p| = |q| = 1$ keeps $|B/C| = \varphi$, while a unitary F with $A = \varphi^{-1}$ has $|B|^2 = 1 - \varphi^{-2} = |C|^2$. So no gauge with $|u| = 1$ makes them unitary; one with $|u| \neq 1$ does ($q^2/p = \varphi^{-1/2}$), which is why Theorem 6.4 allows any $u \neq 0$

and guarantees unitarity when $|u| = 1$ (a sufficient condition: with $p = 4$, $q = 2$ nothing changes at all).

(ii) The factor on $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ is $u_\sigma^{\sigma 1}u_1^{\sigma\sigma}/(u_1^{\sigma\sigma}u_\sigma^{1\sigma}) = 1$. With $u_\psi^{\sigma\sigma} = i$ and the others 1, the factors on $[F_\sigma^{\sigma\sigma\sigma}]_{ef}$ are: $(1,\psi)$: $u_\sigma^{\sigma\psi}u_\psi^{\sigma\sigma}/(u_1^{\sigma\sigma}u_\sigma^{1\sigma}) = i$; $(\psi,1)$: $u_\sigma^{\sigma 1}u_1^{\sigma\sigma}/(u_\psi^{\sigma\sigma}u_\sigma^{\psi\sigma}) = 1/i = -i$; (ψ,ψ): $u_\sigma^{\sigma\psi}u_\psi^{\sigma\sigma}/(u_\psi^{\sigma\sigma}u_\sigma^{\psi\sigma}) = 1$. So κH becomes $\frac{\kappa}{\sqrt{2}}\begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}$, which is unitary (as $|u| = 1$; directly, its columns are orthonormal). On $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}$ the factor is $u_\psi^{\sigma\sigma}u_\sigma^{\psi\sigma}/(u_\sigma^{\sigma\psi}u_\psi^{\sigma\sigma}) = 1$: it stays -1. The printed off-diagonal entries of κH are a choice of gauge; κ is not.

Exercise 4 (Pentagon instances by hand). (i) Write out the pentagon equation of Theorem 6.3 for Fibonacci with $a = b = c = d = e = g = k = \tau$ and $f = l = 1$, and show that it is $A^2 + BCD = 0$. Draw the five trees of the pentagon for these labels.

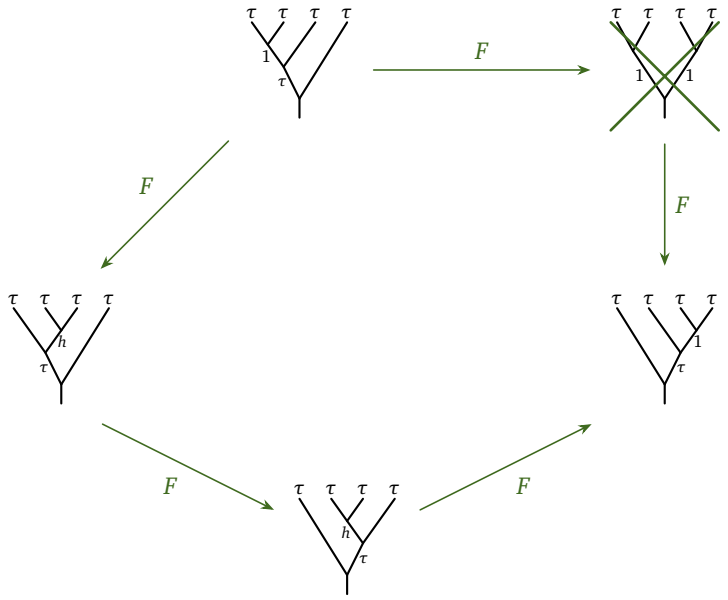
(ii) Find labels whose instance is $BD = xBD$.

(iii) For Ising, with $a = b = c = d = g = k = \sigma$ and $e = f = l = 1$, show that the instance reads

$$1 = \tfrac{\kappa^2}{2}\left(1 + [F_1^{\sigma\psi\sigma}]_{\sigma\sigma}\right)$$

when $F_\sigma^{\sigma\sigma\sigma} = \kappa H$, and check it for the data of Theorem 6.7.

Solution. (i) The equation is $[F_e^{fcd}]_{gl}[F_e^{abl}]_{fk} = \sum_h [F_g^{abc}]_{fh}[F_e^{ahd}]_{gk}[F_k^{bcd}]_{hl}$. On the left, $[F_\tau^{1\tau\tau}]_{\tau 1}$ needs a right tree with $b \otimes c = \tau \otimes \tau \to 1$ and then $1 \otimes 1 \to \tau$, which does not exist, so the left side is 0. On the right, $h = 1$ gives $[F_\tau^{\tau\tau\tau}]_{11}[F_\tau^{\tau 1\tau}]_{\tau\tau}[F_\tau^{\tau\tau\tau}]_{11} = A \cdot 1 \cdot A$ and $h = \tau$ gives $[F_\tau^{\tau\tau\tau}]_{1\tau}[F_\tau^{\tau\tau\tau}]_{\tau\tau}[F_\tau^{\tau\tau\tau}]_{\tau 1} = BDC$. So $0 = A^2 + BCD$. The five trees are those of the pentagon in § 6.2, every leaf τ and the root τ:



Top left is $((ab)c)d$ with $f = 1$, $g = \tau$; top right $(ab)(cd)$ with $f = l = 1$, crossed out because its root vertex $1 \otimes 1 \to \tau$ does not exist; right $a(b(cd))$ with $l = 1$, $k = \tau$; bottom $a((bc)d)$ with $h, k = \tau$; left $(a(bc))d$ with $h, g = \tau$, where $h \in \{1, \tau\}$ is summed. The upper path passes through the crossed-out tree: that is why the left side vanishes.

(ii) With $a = b = c = d = e = f = k = l = \tau$ and $g = 1$: the left side is $[F_\tau^{\tau\tau\tau}]_{1\tau}[F_\tau^{\tau\tau\tau}]_{\tau\tau} = BD$. On the right, $[F_1^{\tau\tau\tau}]_{\tau h}$ is non-zero only for $h = \tau$ (it is x), then $[F_\tau^{\tau\tau\tau}]_{1\tau} = B$ and $[F_\tau^{\tau\tau\tau}]_{\tau\tau} = D$: the right side is xBD.

(iii) Write m_{ef} for the entries of $F_\sigma^{\sigma\sigma\sigma}$. The left side is $[F_1^{1\sigma\sigma}]_{\sigma 1}[F_1^{\sigma\sigma 1}]_{1\sigma} = 1$ (unit legs). On the right, $h = 1$ gives $m_{11}[F_1^{\sigma 1\sigma}]_{\sigma\sigma}m_{11} = m_{11}^2$ and $h = \psi$ gives $m_{1\psi}[F_1^{\sigma\psi\sigma}]_{\sigma\sigma}m_{\psi 1}$. With

$F_\sigma^{\sigma\sigma\sigma} = \kappa H$, $m_{11}^2 = m_{1\psi} m_{\psi 1} = \kappa^2/2$, so the instance is $1 = \frac{\kappa^2}{2}(1 + [F_1^{\sigma\psi\sigma}]_{\sigma\sigma})$. In the data of Theorem 6.7, $[F_1^{\sigma\psi\sigma}]_{\sigma\sigma} = 1$ and $\kappa^2 = 1$: it holds.

Exercise 5 (SU(2) recoupling is a $6j$ symbol). In $\mathrm{Rep}(\mathrm{SU}(2))$ (semisimple, rigid and multiplicity-free, though not finite) the F-symbols are defined as in Definition 6.1. Take isometric vertices given by the Clebsch–Gordan coefficients:

$$\psi_0^{\frac12\frac12}(1) = \tfrac{1}{\sqrt2}(\uparrow\downarrow - \downarrow\uparrow),$$
$$\psi_1^{\frac12\frac12}\colon\ |1,1\rangle \mapsto \uparrow\uparrow,\ |1,0\rangle \mapsto \tfrac{1}{\sqrt2}(\uparrow\downarrow + \downarrow\uparrow),$$
$$\psi_{\frac12}^{1\frac12}\colon\ \uparrow \mapsto \sqrt{\tfrac23}\,|1,1\rangle\downarrow - \sqrt{\tfrac13}\,|1,0\rangle\uparrow,$$
$$\psi_{\frac12}^{\frac121}\colon\ \uparrow \mapsto \sqrt{\tfrac13}\uparrow|1,0\rangle - \sqrt{\tfrac23}\downarrow|1,1\rangle,$$

and $\psi_{\frac12}^{0\frac12} = \psi_{\frac12}^{\frac120} = \mathrm{id}$.

(i) Explain why rescaling the vertices to the normalisation Equation (5.1) does not change the F-symbols. Compute the trees L_0, L_1, R_0, R_1 for three spin-$\frac12$ with total spin $\frac12$ on the state $\uparrow$, and the matrix $F_{\frac12}^{\frac12\frac12\frac12}$.

(ii) Compare with the Racah–Wigner formula

$$[F_d^{abc}]_{ef} = (-1)^{a+b+c+d}\sqrt{(2e+1)(2f+1)}\begin{Bmatrix} a & b & e \\ c & d & f \end{Bmatrix},$$

given that the $6j$ symbols $\left\{\begin{smallmatrix} 1/2 & 1/2 & e \\ 1/2 & 1/2 & f \end{smallmatrix}\right\}$ are $-\frac12$, $\frac12$, $\frac12$, $\frac16$ for $(e,f) = (0,0)$, $(0,1)$, $(1,0)$, $(1,1)$. Compute $d_{1/2}[F_{\frac12}^{\frac12\frac12\frac12}]_{00}$ with $d_{1/2} = 2$.

(iii) Write the pentagon instances with $a = b = c = d = \frac12$, $e = 0$, $g = k = \frac12$ (one for each f, l). Given that the 1×1 symbols in them are 1, show that together they say $(F_{\frac12}^{\frac12\frac12\frac12})^2 = 1$. (This instance is degenerate: (i) already implies it. The general pentagon for these F-symbols is the Biedenharn–Elliott identity, which the check script verifies on representations built with rep-engine.)

Solution. (i) An isometric vertex v and the vertex $\psi = (d_a d_b/d_c)^{1/4} v$ of Equation (5.1) differ by a positive factor, so L_e is multiplied by $(d_a d_b/d_e)^{1/4}(d_e d_c/d_d)^{1/4} = (d_a d_b d_c/d_d)^{1/4}$, the same for every e, and so is R_f. Both sides of $\alpha \circ L_e = \sum_f [F_d^{abc}]_{ef} R_f$ scale alike, and F does not change. In $\mathrm{Vect}_{\mathrm{fd}}$ the associator is the identity on vectors, so on the top state $F_{ef} = \langle R_f, L_e\rangle$ (both families are orthonormal). The trees on $\uparrow$ are

$$L_0 = \tfrac{1}{\sqrt2}(\uparrow\downarrow\uparrow - \downarrow\uparrow\uparrow),$$
$$L_1 = \sqrt{\tfrac23}\,\uparrow\uparrow\downarrow - \tfrac{1}{\sqrt6}(\uparrow\downarrow\uparrow + \downarrow\uparrow\uparrow),$$
$$R_0 = \tfrac{1}{\sqrt2}(\uparrow\uparrow\downarrow - \uparrow\downarrow\uparrow),$$
$$R_1 = \tfrac{1}{\sqrt6}(\uparrow\uparrow\downarrow + \uparrow\downarrow\uparrow) - \sqrt{\tfrac23}\,\downarrow\uparrow\uparrow,$$

(L_e fuses the first two spins to e, then e with the third; R_f fuses the last two to f). The inner products are $\langle R_0, L_0\rangle = -\frac12$, $\langle R_1, L_0\rangle = \frac{1}{\sqrt{12}} + \frac{1}{\sqrt3} = \frac{\sqrt3}{2}$, $\langle R_0, L_1\rangle = \frac{1}{\sqrt2}(\sqrt{\frac23} + \frac{1}{\sqrt6}) = \frac{\sqrt3}{2}$ and

$\langle R_1, L_1\rangle = \frac{1}{3} - \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$:

$$F^{\frac12\frac12\frac12}_{\frac12} = \begin{pmatrix} -\frac12 & \frac{\sqrt3}{2} \\ \frac{\sqrt3}{2} & \frac12 \end{pmatrix},$$

a real orthogonal matrix, as Proposition 6.2 requires.

(ii) Here $(-1)^{a+b+c+d} = (-1)^2 = 1$, and $\sqrt{(2e+1)(2f+1)}$ is $1, \sqrt3, \sqrt3, 3$; with the given $6j$ values the formula gives $-\frac12, \frac{\sqrt3}{2}, \frac{\sqrt3}{2}, \frac12$: the same matrix. And $d_{1/2}[F^{\frac12\frac12\frac12}_{\frac12}]_{00} = 2\cdot(-\frac12) = -1$, which by Equation (6.4) is the indicator $\kappa_{1/2} = -1$ of Lecture 4. The Racah–Wigner formula holds for every label; the pentagon of Theorem 6.3, written in $6j$ symbols, is the Biedenharn–Elliott identity. `scripts/check-sheet6.py` builds the spin representations with rep-engine, computes the F-symbols from Condon–Shortley intertwiners, compares them with the formula up to spin $\frac32$, and checks every pentagon instance with $a, b, c, d \in \{\frac12, 1\}$.

(iii) With $e = 0$ the labels are forced: $g = k = \frac12$, and $f, l, h \in \{0, 1\}$; the left side needs $l = f$ (the tree $f \otimes l \to 0$ exists only for $l = f$). It reads

$$[F_0^{f\frac12\frac12}]_{\frac12 l}\,[F_0^{\frac12\frac12 l}]_{f\frac12} = \sum_h [F^{\frac12\frac12\frac12}_{\frac12}]_{fh}\,[F_0^{\frac12 h\frac12}]_{\frac12\frac12}\,[F^{\frac12\frac12\frac12}_{\frac12}]_{hl}.$$

With the 1×1 symbols equal to 1 this is $\delta_{fl} = \sum_h F_{fh}F_{hl}$: $F^2 = 1$. And indeed $\left(\begin{smallmatrix} -1/2 & \sqrt3/2 \\ \sqrt3/2 & 1/2 \end{smallmatrix}\right)^2$ has diagonal entries $\frac14 + \frac34 = 1$ and off-diagonal entries $-\frac{\sqrt3}{4} + \frac{\sqrt3}{4} = 0$.

Exercise 6 ⋆. *The Ising F-symbols, and why there are no others.* Name the Ising F-symbols without a unit leg as follows:

$$\begin{gathered} n_1 = [F_\psi^{\psi\psi\psi}]_{11},\ n_2 = [F_\sigma^{\psi\psi\sigma}]_{1\sigma}, \\ n_3 = [F_1^{\psi\sigma\sigma}]_{\sigma\psi},\ n_4 = [F_\psi^{\psi\sigma\sigma}]_{\sigma 1}, \\ n_5 = [F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma},\ n_6 = [F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma},\ n_7 = [F_1^{\sigma\sigma\psi}]_{\psi\sigma}, \\ \gamma_1 = [F_1^{\sigma\psi\sigma}]_{\sigma\sigma},\ \gamma_2 = [F_\sigma^{\sigma\psi\psi}]_{\sigma 1}, \\ \gamma_3 = [F_\psi^{\sigma\sigma\psi}]_{1\sigma}, \end{gathered}$$

and $F_\sigma^{\sigma\sigma\sigma} = (m_{ef})$ with $e, f \in \{1, \psi\}$.

(i) Compute the gauge factors of $\gamma_1, \gamma_2, \gamma_3$, and show that every solution of the pentagon with invertible F-matrices (as for any category) is gauge equivalent to one with $\gamma_1 = \gamma_2 = \gamma_3 = 1$.

(ii) With $\gamma_1 = \gamma_2 = \gamma_3 = 1$ the following are pentagon instances:

$$\begin{gathered} n_1 = n_7, \quad n_1 n_2 = n_2, \quad n_2 = 1, \\ n_6^2 = n_2, \quad n_2 n_5^2 = n_2, \quad n_1 = n_2 n_3 n_4, \quad n_3 n_7 = n_4, \\ m_{1\psi} = m_{11}, \quad m_{\psi 1} = n_4 m_{11}, \\ m_{1\psi}m_{\psi 1} + m_{11}^2 = 1, \quad n_6 m_{1\psi}m_{\psi 1} + m_{11}^2 = 0, \\ n_5 m_{\psi\psi} = n_4 m_{1\psi}, \quad n_5 m_{\psi 1} = n_7 m_{\psi\psi}, \\ m_{1\psi}m_{\psi\psi} + m_{11}m_{1\psi} = 0. \end{gathered}$$

Derive the last one from the labels $a = b = c = d = g = k = \sigma$, $e = f = 1$, $l = \psi$. Then show that these fourteen equations have exactly two solutions: $F_\sigma^{\sigma\sigma\sigma} = \kappa H$ with $\kappa = \pm1$, $n_5 = n_6 = -1$, and every other symbol 1.

(iii) Conclude that the Ising fusion ring carries exactly two fusion categories up to gauge, and that they are told apart by κ.

Solution. (i) By Theorem 6.4, $[F_d^{abc}]_{ef}$ has the factor $u_d^{af}u_f^{bc}/(u_e^{ab}u_d^{ec})$, with $u = 1$ on unit legs. For γ_1: $u_1^{\sigma\sigma}u_\sigma^{\psi\sigma}/(u_\sigma^{\sigma\psi}u_1^{\sigma\sigma}) = u_\sigma^{\psi\sigma}/u_\sigma^{\sigma\psi}$. For γ_2: $u_\sigma^{\sigma 1}u_1^{\psi\psi}/(u_\sigma^{\sigma\psi}u_\sigma^{\sigma\psi}) = u_1^{\psi\psi}/(u_\sigma^{\sigma\psi})^2$. For γ_3: $u_\psi^{\sigma\sigma}u_\sigma^{\sigma\psi}/(u_1^{\sigma\sigma}u_\psi^{1\psi}) = u_\sigma^{\sigma\psi}u_\psi^{\sigma\sigma}/u_1^{\sigma\sigma}$. Each γ_i is a 1×1 block of an invertible F-matrix, so it is non-zero. Take $u_\sigma^{\sigma\psi} = u_1^{\sigma\sigma} = 1$, $u_\sigma^{\psi\sigma} = 1/\gamma_1$, $u_1^{\psi\psi} = 1/\gamma_2$ and $u_\psi^{\sigma\sigma} = 1/\gamma_3$: the new values are all 1.

(ii) The labels $a = b = c = d = g = k = \sigma$, $e = f = 1$, $l = \psi$ give, on the left, $[F_1^{1\sigma\sigma}]_{\sigma\psi}[F_1^{\sigma\sigma\psi}]_{1\sigma}$, which is 0: the first needs a right tree with $\sigma\otimes\sigma\to\psi$ and $1\otimes\psi\to 1$. On the right, $h = 1$ gives $m_{11}[F_1^{\sigma 1\sigma}]_{\sigma\sigma}m_{1\psi} = m_{11}m_{1\psi}$ and $h = \psi$ gives $m_{1\psi}\,\gamma_1\,m_{\psi\psi}$; with $\gamma_1 = 1$ this is $0 = m_{11}m_{1\psi} + m_{1\psi}m_{\psi\psi}$, the last equation.

Now solve. $n_2 = 1$, so $n_1n_2 = n_2$ gives $n_1 = 1$, and $n_1 = n_7$ gives $n_7 = 1$; $n_6^2 = n_5^2 = 1$. With $m_{1\psi} = m_{11}$, the equation $m_{1\psi}m_{\psi 1} + m_{11}^2 = 1$ reads $m_{11}m_{\psi 1} + m_{11}^2 = 1$, so $m_{11}\neq 0$; the last equation then gives $m_{\psi\psi} = -m_{11}$. Substituting $m_{\psi 1} = n_4m_{11}$: the same equation becomes $(1+n_4)m_{11}^2 = 1$, so $n_4\neq -1$, and $n_6m_{1\psi}m_{\psi 1} + m_{11}^2 = 0$ becomes $(n_6n_4+1)m_{11}^2 = 0$, so $n_6n_4 = -1$. From $n_1 = n_2n_3n_4$, $n_3n_4 = 1$; from $n_3n_7 = n_4$, $n_3 = n_4$; so $n_4^2 = 1$, and as $n_4\neq -1$, $n_4 = 1 = n_3$. Then $n_6 = -1$ and $2m_{11}^2 = 1$, so $m_{11} = \kappa/\sqrt{2}$ with $\kappa = \pm 1$, and $m_{\psi 1} = m_{11}$. The twelfth equation, $n_5(-m_{11}) = m_{11}$, gives $n_5 = -1$, and the thirteenth, $-m_{11} = -m_{11}$, holds. So $F_\sigma^{\sigma\sigma\sigma} = \frac{\kappa}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right) = \kappa H$, $n_5 = [F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma} = -1$, $n_6 = [F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma} = -1$ and all the others are 1: for each κ exactly one solution of the fourteen equations, which is the data of Theorem 6.7. It solves the whole pentagon (the lecture's verification, repeated by the script).

(iii) By (i) every solution of the pentagon is gauge equivalent to one with $\gamma_1 = \gamma_2 = \gamma_3 = 1$; by (ii) such a solution is one of the two. They are not gauge equivalent, since $d_\sigma[F_\sigma^{\sigma\sigma\sigma}]_{11} = \sqrt{2}\cdot\kappa/\sqrt{2} = \kappa$ is invariant (Equation (6.4)). So up to gauge there are exactly two. Each defines a monoidal category by the converse in Theorem 6.3 (sketched there): the course's Ising ($\kappa = +1$) and the category of $\mathrm{SU}(2)_2$ ($\kappa = -1$). This is the Tambara–Yamagami classification for $\mathbb{Z}_2$, which Lecture 6 cites (EGNO Prop. 9.14.9), in F-symbols; that two gauge-inequivalent solutions give inequivalent categories is the lecture's sketch.